

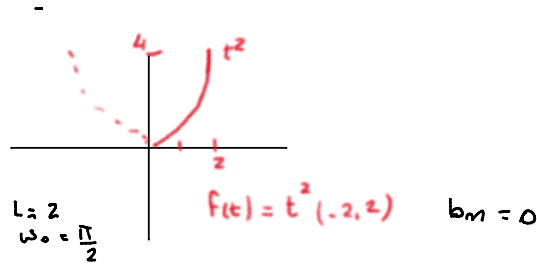
Resolucion 1 parcial tema 2 2024

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Ejercicio 1

Dada la función $f(t) = t^2, t \in (0, 2)$.

- Completar la función para que la STF sea solo de cosenos y desarrollarla.
- Con el resultado obtenido en el punto a) hallar el valor al cual converge $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$
- Dada la función $f(t) = t^2$ definida en el intervalo $(0, 2)$. Completarla analítica y gráficamente para que sea impar y tenga simetría de media onda.



$$a) \quad a_0 = \frac{2}{L} \int_0^2 t^2 dt = \frac{2}{2} \int_0^2 t^2 dt = \left. \frac{t^3}{3} \right|_0^2 = \frac{8}{3} \quad \frac{a_0}{2} = \frac{4}{3}$$

$$a_m = \frac{2}{L} \int_0^2 t^2 \cos\left(m \frac{\pi}{2} t\right) dt = \int_0^2 t^2 \cos\left(m \frac{\pi}{2} t\right) dt = \left(\frac{2t \cos\left(m \frac{\pi}{2} t\right)}{m^2 \frac{\pi^2}{4}} + \left(\frac{t^2}{m \frac{\pi}{2}} - \frac{2}{m^3 \frac{\pi^3}{8}} \right) \sin\left(m \frac{\pi}{2} t\right) \right) \Big|_0^2$$

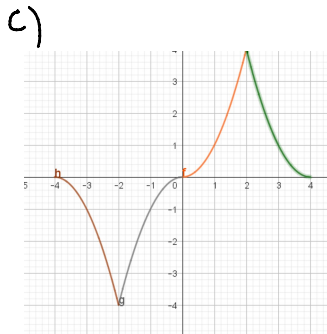
$$a_m = \frac{4(-1)^m}{m^2 \cdot \frac{\pi^2}{4}} \Rightarrow a_m = \frac{16(-1)^m}{m^2 \cdot \pi^2} \quad S(x) = \frac{4}{3} + \sum_{m=1}^{\infty} \frac{16(-1)^m}{m^2 \pi^2} \cos\left(m \frac{\pi}{2} t\right)$$

$$b) \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = ? \quad S(0) = \frac{4}{3} + \sum_{n=1}^{\infty} \frac{16(-1)^n}{n^2 \pi^2} = 0$$

$$\frac{4}{3} + \frac{16}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = 0$$

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{4}{3} \cdot \frac{\pi^2}{16} = -\frac{\pi^2}{12}$$

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$



$$f(t) = \begin{cases} t^2 & (0, 2) \\ -t^2 & (-2, 0) \\ (t-4)^2 & (2, 4) \\ -(t+4)^2 & (-4, -2) \end{cases}$$

Ejercicio 2

Resolver usando Transformada de Laplace

$$y(t) - 2 \int_0^t y(u) \cos(t-u) du = 1$$

$$\mathcal{L}\{y(t)\} - 2 \mathcal{L}\left\{\int_0^t y(u) \cos(t-u) du\right\} = \mathcal{L}\{1\}$$

$$Y(s) - 2 Y(s) \frac{s}{s^2 + 1} = \frac{1}{s}$$

$$Y(s) \left(1 - \frac{2s}{s^2 + 1}\right) = \frac{1}{s} \Rightarrow Y(s) \frac{s^2 + 1 - 2s}{s^2 + 1} = \frac{1}{s}$$

$$Y(s) = \frac{s^2 + 1}{s(s^2 - 2s + 1)} = \frac{A}{s} + \frac{B}{(s-1)^2} + \frac{C}{s-1}$$

$$s^2 = A(s^2 - 2s + 1) + B(s-1) + C(s-1)^2$$

$$s^2 = As^2 - 2As + A + Bs - B + Cs^2 - 2Cs + C$$

$$s^2 = (A+C)s^2 + (-2A+B-2C)s + (A-B+C)$$

$$1 = A+C \quad -2A+B-2C = 0 \quad A-B+C = 1$$

$$A = 1 \quad C = 0 \quad B = 2$$

$$Y(s) = \frac{1}{s} + \frac{2}{(s-1)^2}$$

$$Y(s) = \frac{s^2+1}{s(s-1)^2} = \frac{A}{s} + \frac{B}{(s-1)^2} + \frac{C}{s-1}$$

$$\rightarrow A(s^2-2s+1) + Bs + C(s-1) = s^2+1$$

$$s^2 - 2As + 1 + Bs + Cs - C = s^2 + 1$$

$$1 - C = 1 \rightarrow C = 0$$

$$Y(s) = \frac{1}{s} + \frac{2}{(s-1)^2} \rightarrow y(t) = 1 + 2te^t$$

Ejercicio 3

Dado los siguientes conjuntos:

$$A = \left\{ z \in \mathbb{C} : |z|^2 + 4\bar{z} = \frac{33}{4} - 4 \operatorname{Im}(z)j \right\}$$

$$B = \left\{ z \in \mathbb{C} : \sqrt{\frac{11}{2} - \frac{1}{2}j} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \right\}$$

- Dibujar los conjuntos A y B en un mismo gráfico
- Hallar $A \cap B$.

(Usar siempre valor principal cuando se necesite)

$$A) |z|^2 + 4\bar{z} = \frac{33}{4} - 4 \operatorname{Im}(z)j$$

$$x^2 + y^2 + 4x - 4yj = \frac{33}{4} - 4yj \rightarrow x^2 + 4x + y^2 = \frac{33}{4}$$

$$x^2 + 4x + 4 + y^2 = \frac{33}{4} + 4$$

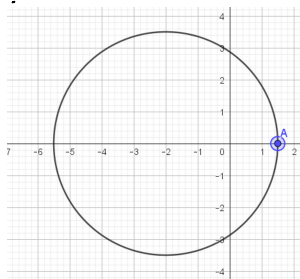
$$(x+2)^2 + y^2 = \frac{49}{4}$$

$$B) \sqrt{\frac{11}{2} - \frac{1}{2}j} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j \rightarrow \frac{11}{2} \ln\left(\frac{\sqrt{2}}{2} - \frac{1}{2}j\right) = \ln\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j\right)$$

$$\downarrow$$

$$\operatorname{tg}^{-1}\left(-\frac{1}{2} \cdot \frac{2}{\sqrt{3}}\right) = -\frac{\pi}{6} + 2\pi$$

$$\frac{z}{x} \cdot \frac{x}{\sqrt{3}} = \frac{x}{\frac{11}{2}} \rightarrow z = \frac{3}{2}$$



$$A \cap B = \left\{ \left(\frac{3}{2}, 0 \right) \right\}$$

Ejercicio 4

Dada la función de transferencia $G(s) = \frac{5(s^2 - ks - 2)}{(s^2 + s - 6)(s^2 + 2s + 2)}$

- Hallar k para que el sistema sea estable.
- Con el k hallado, obtener la respuesta temporal del sistema a una entrada $f(t) = \delta(t)$
- Decir si la siguiente afirmación es verdadera o falsa: El valor de $|G(2+j)| = \frac{5\sqrt{5}}{39}$. (Utilice el método gráfico)

$$a) s^2 + s - 6 = (s-2)(s+3) \rightarrow s^2 - ks - 2 \text{ debe ser } 0$$

$$4 - 2k - 2 = 0 \Rightarrow k = 1$$

$$s^2 - 2s - 2 = (s+1)(s-2)$$

$$G(s) = \frac{5(s+1)(s-2)}{(s-2)(s+3)(s^2+2s+2)} \quad \text{SE}$$

$$b) f(t) = \delta(t) \rightarrow F(s) = 1 \quad Y(s) = F(s) \cdot G(s) = 1 \cdot \frac{5(s+1)}{(s+3)(s^2+2s+2)}$$

$$Y(s) = \frac{A}{s+3} + \frac{Bs+C}{s^2+2s+2} \quad \therefore \begin{aligned} -2(A+2B+C) + (Bs+C)(s+3) &= 5s+5 \\ -2+0 &= 0 \rightarrow B=2 \\ -4+3C &= 5 \quad C=3 \end{aligned}$$

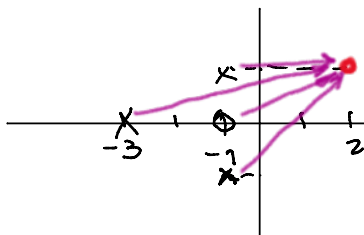
$$Y(s) = \frac{-2}{s+3} + \frac{2(s+1)}{(s+1)^2+1} + \frac{3}{(s+1)^2+1}$$

$$Y(s) = \frac{-2}{s+3} + \frac{2(s+1)}{(s+1)^2+1} + \frac{1}{(s+1)^2+1}$$

$$y(t) = -2e^{-3t} + 2\cos(t)e^{-t} + \sin(t)e^{-t}$$

$$3) |G(z+j)|$$

(V)



$$|G(z+j)| = \frac{5\sqrt{10}}{\sqrt{26} \cdot 3 \cdot \sqrt{13}}$$

$$|G(z+j)| = \frac{5\sqrt{2} \cdot \sqrt{5}}{\sqrt{2} \cdot \sqrt{13} \cdot 3\sqrt{13}}$$

$$|G(z+j)| = \frac{5\sqrt{5}}{39}$$

Ejercicio 5

Resolver la ecuación en diferencia $x(n+2) - 4x(n) = 3^n$ $x(0) = 1, x(1) = 2$

$$x(n+2) - 4x(n) = 3^n \quad x(0) = 1 \quad x(1) = 2$$

$$z^2 X(z) - z^2 - 2z - 4X(z) = \frac{z}{z-3}$$

$$X(z)(z^2 - 4) = \frac{z}{z-3} + z^2 + 2z \quad z^2 - z - 5$$

$$X(z) = \frac{z + z^2(z-3) + 2z(z-3)}{(z-3)(z+2)(z-2)} = z \left(\frac{1 + z(z-3) + 2(z-3)}{(z-3)(z+2)(z-2)} \right)$$

$$X(z) = z \left(\frac{A}{z-3} + \frac{B}{z+2} + \frac{C}{z-2} \right) = \frac{1}{5} \frac{z}{z-3} + \frac{1}{20} \frac{z}{z+2} + \frac{3}{4} \frac{z}{z-2}$$

$$x(t) = \frac{1}{5} \cdot 3^n + \frac{1}{20} (-2)^n + \frac{3}{4} \cdot 2^n$$

$$x(1) = 2 \quad \frac{1}{5} \cdot 3 - \frac{1}{10} + \frac{3}{2} = \frac{3}{5} - \frac{1}{10} + \frac{3}{2} = \frac{6-1+15}{10} = \frac{20}{10} = 2$$

VERIFIED!